

COVID-19 GLOBAL ANALYTICS REPORT

Power BI Data Preparation, Data Modelling and DAX Analysis

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Dataset downloaded: 2 March 2026

RESEARCH QUESTIONS

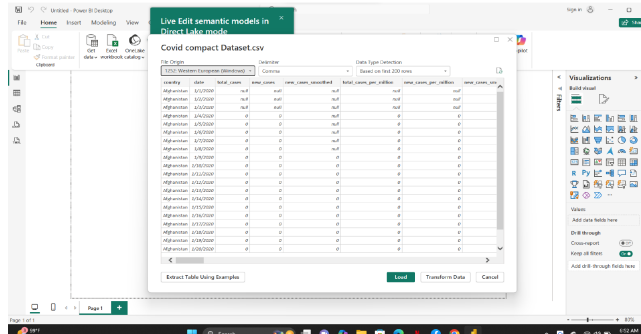
1. What is the global daily pattern of COVID-19 cases over time?
2. How does the global stringency index change over time?
3. What is the relationship between total cases per million people and total deaths per million people?
4. Which country has the highest cumulative number of cases?
5. How does infection rate vary between countries and continents with different population sizes?
6. Is there a visible relationship between continent, GDP per capita and cumulative case fatality rate?
7. How did the top five infected countries change in rank across different quarters of the pandemic?
8. What is the average case fatality rate (CFR%) across the world?
9. Which five countries recorded the highest total COVID-19 cases from 2020 to 2025?
10. Which continent reached the highest peak in smoothed new cases?

The report below summarizes the same workflow shown in the original Power BI project, but uses shorter sentences and simpler explanations.

1. DATA LOADING AND CLEANING

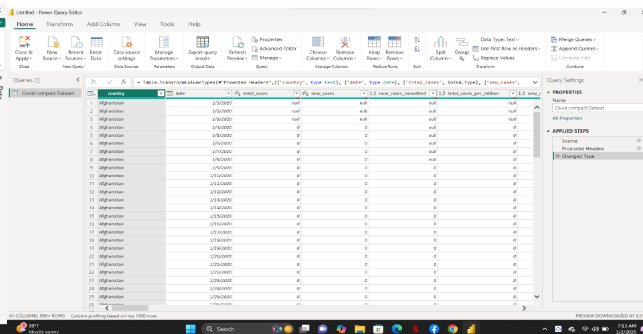
The COVID-19 dataset was first brought into Power Query. The data was checked, loaded, and cleaned before it was used for analysis. Cleaning helps make the data easier to work with and reduces problems during modelling and calculation.

• LOADING/TRANSFORMING THE DATA INTO POWERQUERY.



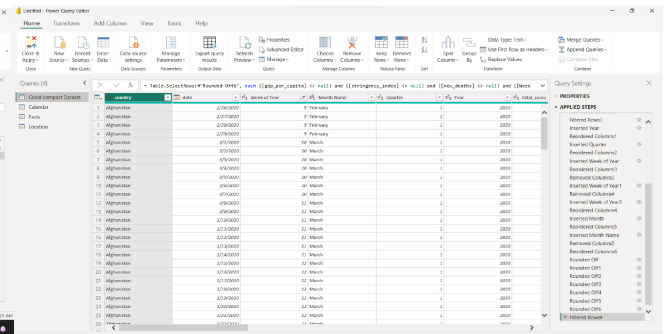
Data loaded into Power Query

• LOADED DATA



Initial data view after loading

• CLEANED DATA.



Cleaned data after preparation

2. TABLE PREPARATION

The dataset was divided into useful tables. A Location table was prepared for country and continent details. A Calendar table was created for date analysis. A Facts table was used for the main COVID-19 records and measures.

- LOCATION TABLE CREATION

[illegible]

Location table creation

- CALENDAR TABLE CREATION

[illegible]

Calendar table creation

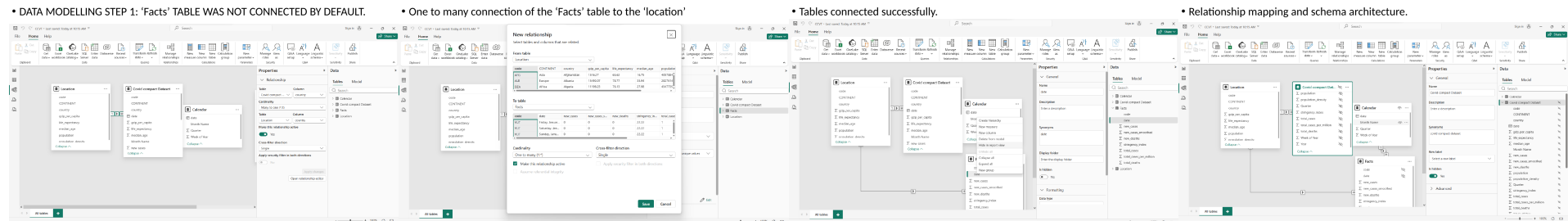
- FACTS TABLE CREATION

[illegible]

Facts table creation

3. DATA MODELLING AND RELATIONSHIPS

The Facts table was not connected to the other tables automatically, so the relationships were created manually. A one-to-many relationship was made between the Facts table and the Location table. The tables were then connected into one model.



Initial model before the missing relationship was fixed

One-to-many relationship setup

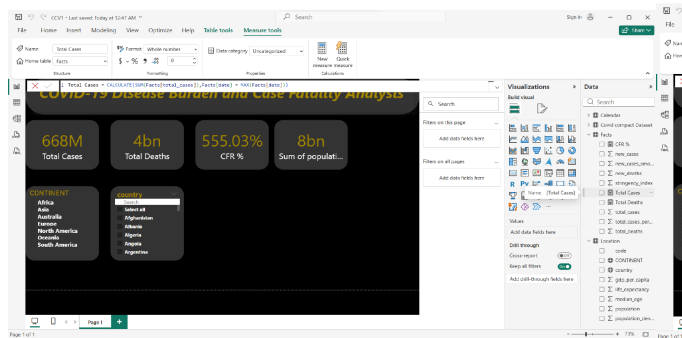
Tables connected successfully

Final relationship map and schema

4. DAX MEASURES AND DASHBOARD VALUES

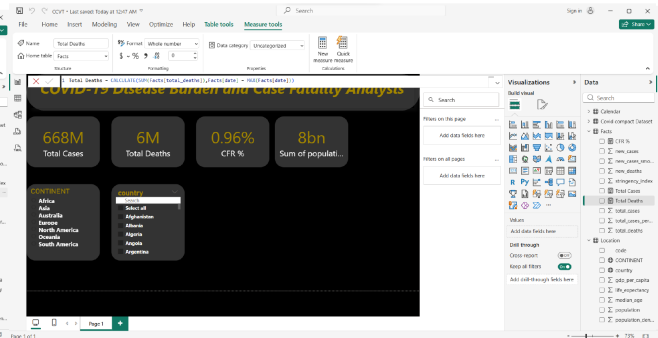
DAX measures were created to produce the main values used in the dashboard. These measures include total cases, total deaths, infection rate, average stringency index, and case fatality rate (CFR%).

• TOTAL CASES MEASURE CREATION



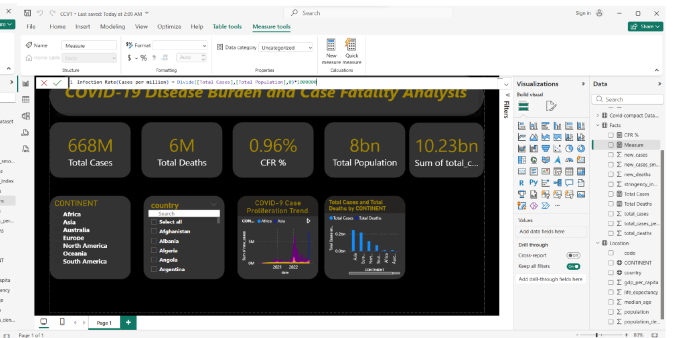
Total Cases measure

• TOTAL DEATHS CALCULATION



Total Deaths measure

• INFECTION RATE CALCULATION

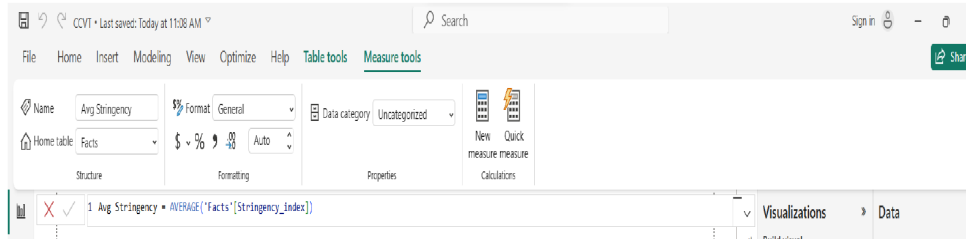


Infection Rate calculation

5. FINAL DAX CALCULATIONS AND SUMMARY

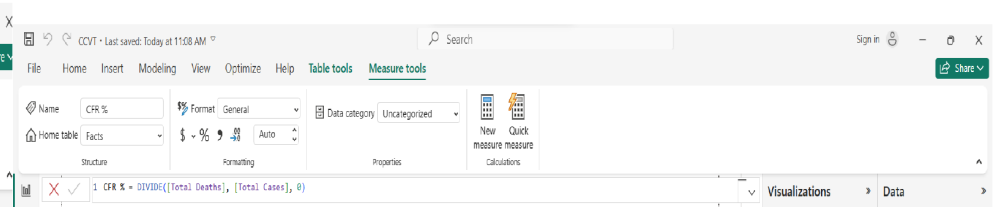
The final calculations support the main research questions and allow the dashboard to compare countries, continents, dates, cases, deaths, population and other indicators. The screenshots below show the remaining calculations used in the project.

- AVERAGE STRINGENCY INDEX CALCULATION



Average Stringency Index calculation

- CFR % CALCULATION



CFR% calculation

PROJECT SUMMARY

The project follows a clear Power BI process: load the COVID-19 data, clean the data, create supporting tables, build relationships, and create DAX measures. This structure makes the dataset ready for dashboard analysis and comparison across time, countries and continents.

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